IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE/EIE PART I: MEng, BEng and ACGI

Corrected Copy

ANALYSIS OF CIRCUITS

Friday, 31 May 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A.S. Holmes
Second Marker(s) : P. Georgiou

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Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.
- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.
- Times are in seconds unless otherwise stated.

Filter transfer function

The standard form for the transfer function of a second order low-pass filter is:

$$H(j\omega) = \frac{1}{\left(\frac{j\omega}{\omega_0}\right)^2 + 2\zeta\left(\frac{j\omega}{\omega_0}\right) + 1}$$

Compound angle formula

The compound angle formula for $\tan(A - B)$ is:

$$\tan(A - B) = \frac{\tan(A) - \tan(B)}{1 + \tan(A)\tan(B)}$$

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The Questions

1. a) Using nodal analysis, calculate the voltages at nodes X and Y of Figure 1.1. [5]

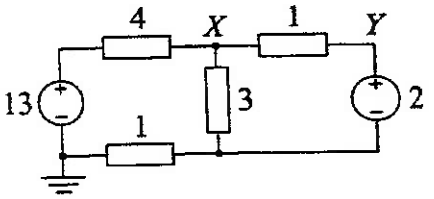


Figure 1.1

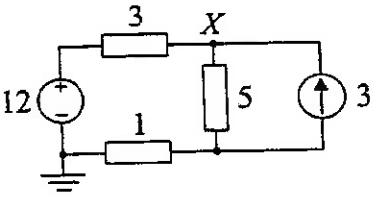


Figure 1.2

b) Using the principle of superposition, calculate the voltage at node X in Figure 1.2. [5]

c) Draw the Thévenin equivalent of the network shown in Figure 1.3 and assign values to its components. [5]

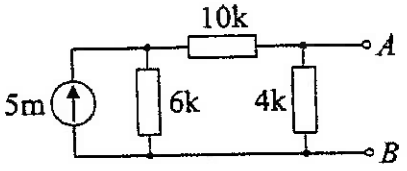


Figure 1.3

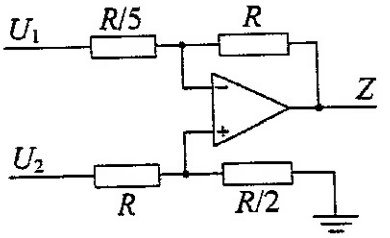


Figure 1.4

d) For the circuit in Figure 1.4, derive an expression relating Z to U_1 and U_2. You should assume the opamp is ideal. [5]

e) For the circuit in Figure 1.5, calculate the complex power associated with each component, including the source. The source has a frequency of 50 Hz. [5]

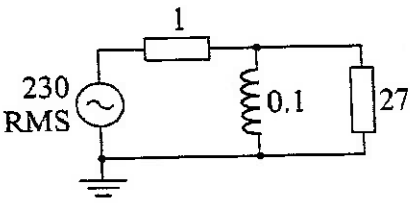


Figure 1.5

Question 1 continues on the next page...

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Question 1 continued

f) Derive the transfer function of the circuit in Figure 1.6. Hence sketch a straight line approximation to the magnitude response, $\left|\frac{Y}{X}(j\omega)\right|$, assigning gain values in dB to the high- and low-frequency asymptotes. Also label the corner frequencies, expressing them in terms of R and C. [6]

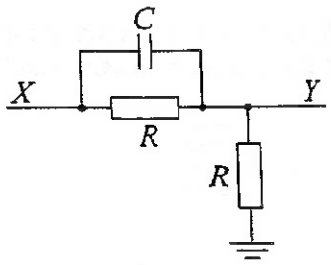


Figure 1.6

g) Determine the gain, $\frac{Y}{X}$, for the block diagram in Figure 1.7. The arrow on each gain block identifies the output of the block. The open circles are adders with signed inputs as indicated. [4]

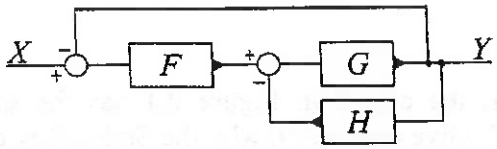


Figure 1.7

h) In the circuit of Figure 1.8, the diode has a forward voltage of 0.7 V when it is conducting but is otherwise ideal. You may assume the opamp is ideal. Draw a dimensioned sketch showing the output waveform y(t) over the time interval $0 \le t \le 3$ ms when $x(t) = 2\sin(2000\pi t)$. [5]

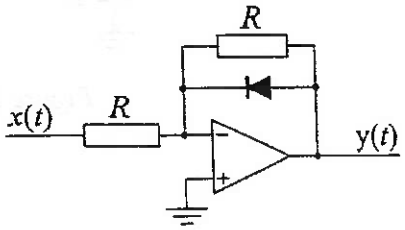


Figure 1.8

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2. a) Figure 2.1 shows a Sallen-Key low-pass filter. Using nodal analysis, and assuming the opamp is ideal, show that the transfer function is: [8]

$$H(j\omega)=\frac{Y}{X}=\frac{1}{mn(j\omega RC)^2+(m+1)j\omega RC+1}$$

b) By comparing the above expression for $H(j\omega)$ with the standard form for the transfer function of a second order filter, show that the corner frequency and quality factor are given by: [6]

$$\omega_0=\frac{1}{\sqrt{mn}RC} \quad ; \quad Q=\frac{\sqrt{mn}}{m+1}$$

c) A circuit of the type in Figure 2.1 is constructed with $m = 1$, $n = 16$, $C = 10$ nF. What value of R should be used if the filter is to have a corner of 1 kHz? [4]

Assuming the above design values, draw a dimensioned sketch showing the magnitude response $|H(j\omega)|$ in dB. Your sketch should show the low- and high-frequency asymptotes and also illustrate the behaviour of the gain around the corner frequency. [6]

d) Show that, in the case $m = n = 1$, the circuit in Figure 2.1 has the same transfer function as the circuit in Figure 2.2. Give one reason why the Sallen-Key circuit might be preferred in this case. [6]

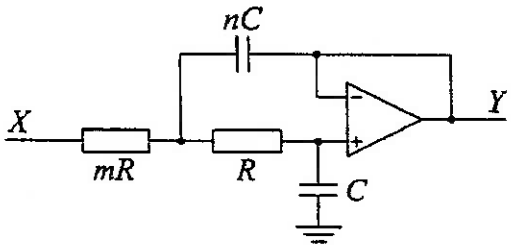


Figure 2.1

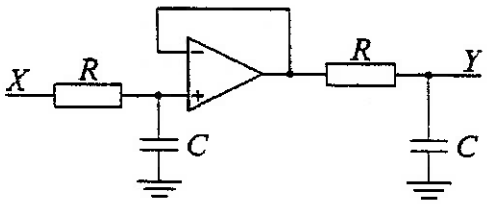


Figure 2.2

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3. This question relates to the circuit in Figure 3.1 which is a first-order low-pass filter with a gain floor. You should assume $k > 1$.

a) Derive the transfer function $H(j\omega)$ of the filter and show that the magnitude of the gain and the phase shift can be expressed as:

$$|H| = \sqrt{\frac{1+u^2}{1+k^2u^2}} \quad ; \quad \angle H = \arctan(u) - \arctan(ku)$$

where $u = \omega RC$. [4]

b) Show that the phase shift passes through a minimum when $u = \frac{1}{\sqrt{k}}$ and that the magnitude and phase of the transfer function at this frequency are:

$$\left|H\left(\frac{j}{\sqrt{k}RC}\right)\right| = \frac{1}{\sqrt{k}} \quad ; \quad \angle H\left(\frac{j}{\sqrt{k}RC}\right) = -\arctan\left(\frac{k-1}{2\sqrt{k}}\right)$$ [8]

c) Draw a dimensioned sketch showing the phase response of a filter having $k = 3$. Your sketch should include both a straight line approximation and a curve that follows more closely the shape of the actual response. [6]

d) An input signal as shown in Figure 3.2(a), with $x(t) = 0$ for $t < 0$ and $x(t) = +5$ for $t \geq 0$, is applied to a filter with $C = 18$ nF, $R = 5.1$ kΩ and $k = 3$. Calculate the steady state outputs for $t < 0$ and $t > 0$. Also, by applying appropriate boundary conditions, calculate the output voltage at $t = 0^+$. Hence derive a full expression for $y(t)$. Also draw a dimensioned sketch of $y(t)$ covering the time interval -1 ms $\leq t \leq 3$ ms. [6]

e) Repeat the steps in part d) to derive a full expression for $y(t)$ in the case where the input voltage is as shown in Figure 3.2(b), with $x(t) = 0$ for $t < 0$ and $x(t) = 5\sin(2000\pi t)$ for $t \geq 0$. You do not need to sketch $y(t)$ for this case. [6]

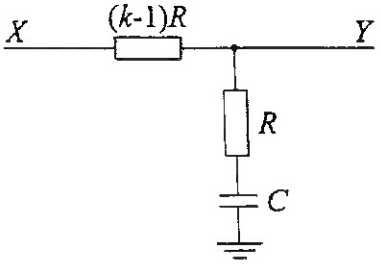


Figure 3.1

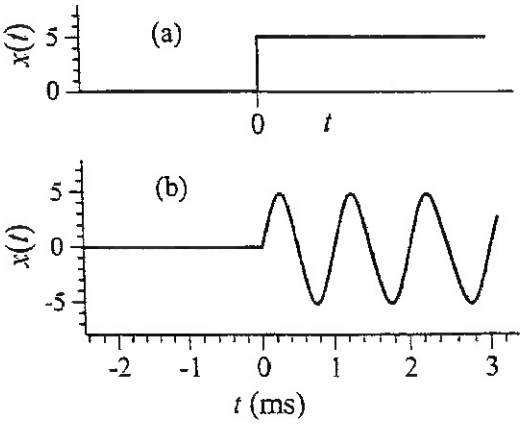


Figure 3.2

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